# Substitutions

As we've seen, first order linear differential equations and first order separable differential equations always have a solution, and we know now how to find their solutions.

First order differential equations that aren't specifically linear or separable may or may not have a solution. To determine whether or not a non-linear, non-separable equation has a solution, one approach we can take is to try transforming the differential equation into a linear or separable equation.

In other words, if we can find a way to rewrite a non-linear, non-separable equation as an actual linear or separable differential equation, then of course we'll be able to find the solution to the equation.

## Making a substitution

That's where substitution comes in. The substitution we want to make, also called a change of variable, is the one that allows us to convert the differential equation into a linear or separable equation.

Given an equation in terms of $y$, $y'$, and $x$, we'll want to find a way to replace all of those with $u$ and $u'$. Sometimes we'll be working with a differential equation in the form

$$y' = F(ax + by)$$

in which case we'll use the specific substitution

$$u = ax + by$$

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$$u' = a + by'$$

Let's do an example to see what a substitution might look like.

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**Example**

Use a change of variable to solve the differential equation.

$$y' = 2x + y$$

If we choose the substitution $u = ax + by$, then we can set up the substitution

$$u = 2x + y$$

$$u' = 2 + y'$$

and then solve this second equation $u' = 2 + y'$ for $y'$ to get $y' = u' - 2$. Then we'll substitute $y' = u' - 2$ into the left side of the original differential equation, and $u = 2x + y$ into the right side of the original differential equation, and we'll get

$$u' - 2 = u$$

Now the equation is separable, so we'll separate variables,

$$\frac{du}{dx} = u + 2$$

$$du = (u + 2)\, dx$$

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$$\frac{1}{u+2}\,du = dx$$

and then integrate both sides.

$$\int \frac{1}{u+2}\,du = \int dx$$

$$\ln(u+2) = x + C$$

$$e^{\ln(u+2)} = e^{x+C}$$

$$u + 2 = Ce^x$$

$$u = Ce^x - 2$$

We'll back-substitute using $u = 2x + y$ and then solve for $y$ to find the solution to the differential equation.

$$2x + y = Ce^x - 2$$

$$y = Ce^x - 2x - 2$$

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Let's do another example with a $u = ax + by$ substitution.

**Example**

Use a change of variable to solve the differential equation.

$$y' - (4x - y - 1)^2 = 0$$

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If we choose the substitution $u = ax + by$, then we can set up the substitution

$$u = 4x - y$$

$$u' = 4 - y'$$

and then solve this second equation $u' = 4 - y'$ for $y'$ to get $y' = 4 - u'$. Then we'll substitute $y' = 4 - u'$ and $u = 4x - y$ into the left side of the original differential equation, and we'll get

$$4 - u' - (u - 1)^2 = 0$$

$$u' = 4 - (u - 1)^2$$

Now the equation is separable, so we'll separate variables,

$$\frac{du}{dx} = 4 - (u - 1)^2$$

$$du = (4 - (u - 1)^2)\, dx$$

$$\frac{du}{4 - (u - 1)^2} = dx$$

and then integrate both sides.

$$\int \frac{du}{4 - (u - 1)^2} = \int dx$$

$$\int \frac{du}{4 - (u^2 - 2u + 1)} = \int dx$$

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$$\int \frac{du}{4 - u^2 + 2u - 1} = \int dx$$

$$\int \frac{du}{3 + 2u - u^2} = \int dx$$

$$-\int \frac{du}{u^2 - 2u - 3} = \int dx$$

$$-\int \frac{du}{(u - 3)(u + 1)} = \int dx$$

Use partial fractions to rewrite the integral on the left,

$$\frac{1}{4}\int \frac{1}{u + 1} - \frac{1}{u - 3} = \int dx$$

then evaluate the integrals on both sides.

$$\frac{1}{4}(\ln|u + 1| - \ln|u - 3|) = x + C$$

$$\ln\left|\frac{u + 1}{u - 3}\right| = 4x + C$$

Raise both sides to the base $e$, then solve for $u$.

$$\frac{u + 1}{u - 3} = Ce^{4x}$$

$$u + 1 = (u - 3)Ce^{4x}$$

$$u + 1 = uCe^{4x} - 3Ce^{4x}$$

$$u - uCe^{4x} = -1 - 3Ce^{4x}$$

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$$u(1 - Ce^{4x}) = -(1 + 3Ce^{4x})$$

$$u = -\frac{1 + 3Ce^{4x}}{1 - Ce^{4x}}$$

Now we can back-substitute $u = 4x - y$ and then solve for $y$ to find the solution to the differential equation.

$$4x - y = -\frac{1 + 3Ce^{4x}}{1 - Ce^{4x}}$$

$$y = 4x + \frac{1 + 3Ce^{4x}}{1 - Ce^{4x}}$$

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